Higher functoriality of truncation: a homotopy induces a homotopy with a specified coherence square

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:trunc-htpy' (source lines 569-578, Lemma 7.3.5); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own construction of the functorial action of truncation together with the naturality homotopy, including the numbered display that the statement refers to.

Editorial note. The retained counter values (chapter 7, section 3, theorem offset 4, equation offset 3) make the retained lemma print as Lemma 7.3.5 and the naturality display as equation (7.3.4), matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: higher functoriality of truncation with its coherence square (Lemma 7.3.5).

Allowed standard prerequisites (editorial).

- The universal property of n-truncation. Precomposition with |.|_n : A -> trunc n A is an equivalence from trunc n A -> B to A -> B whenever B is an n-type.. It is invoked at source line 582 to conclude that the equality of the composites comes from an equality of the functors themselves.
- Functoriality of truncation and its naturality homotopy. The recursion principle turns f : A -> B into trunc n f, and nat^f_n : prd a:A trunc n f(|a|_n) = |f(a)|_n.. The statement is phrased with nat^f_n and nat^g_n, and the proof composes with the paths they provide.
- Function extensionality. A homotopy between two functions gives an equality of the functions.. It converts the homotopy obtained at source line 581 into an equality before the universal property is applied.

In particular, this implies that the n-truncation is functorial: given $f : A \to B$, applying the recursion principle to the composite $A \xrightarrow{f} B \to \|B\|_n$ yields a map $\|f\|_n : \|A\|_n \to \|B\|_n$. By definition, we have a homotopy

$$\mathsf{nat}_n^f : \prod_{a:A} \|f\|_n(|a|_n) = |f(a)|_n, \tag{7.3.4}$$

expressing *naturality* of the maps $|-|_n$.

Uniqueness implies functoriality laws such as $\|g \circ f\|_n = \|g\|_n \circ \|f\|_n$ and $\|\mathsf{id}_A\|_n = \mathsf{id}_{\|A\|_n}$, with attendant coherence laws. We also have higher functoriality, for instance:

Lemma 7.3.5. *Given $f, g : A \to B$ and a homotopy $h : f \sim g$, there is an induced homotopy $\|h\|_n : \|f\|_n \sim \|g\|_n$ such that the composite*

$$|f(a)|_n \xrightarrow{\mathsf{nat}_n^f(a)^{-1}} \|f\|_n(|a|_n) \xrightarrow{\|h\|_n(|a|_n)} \|g\|_n(|a|_n) \xrightarrow{\mathsf{nat}_n^g(a)} |g(a)|_n \quad (7.3.6)$$

is equal to $\mathsf{ap}_{|-|_n}(h(a))$.

Proof. First, we indeed have a homotopy with components $\mathsf{ap}_{|-|_n}(h(a)) : |f(a)|_n = |g(a)|_n$. Composing on either sides with the paths $|f(a)|_n = \|f\|_n(|a|_n)$ and $|g(a)|_n = \|g\|_n(|a|_n)$, which arise from the definitions of $\|f\|_n$ and $\|g\|_n$, we obtain a homotopy $(\|f\|_n \circ |-|_n) \sim (\|g\|_n \circ |-|_n)$, and hence an equality by function extensionality. But since $(- \circ |-|_n)$ is an equivalence, there must be a path $\|f\|_n = \|g\|_n$ inducing it, and the coherence laws for function extensionality imply (7.3.6). □